Oeconomica Micro Theory Cohort: Lecture Notes

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1 Introduction

Aden Chen and Shravan Hari

The aim of this series of lectures is to expose students to a wide range of topics in Microeconomic Theory. We hope to deliver economic insights with minimal mathematical formalism. This means that we sometimes have to sacrifice full generality to present a cleaner story, and mathematical details such as the existence of optimizers and the difference between max and sup will generally be ignored.[1]

Additional notes on economic interpretation, history, and terminology will generally be displayed on the margin.

This set of notes are written under the assumption that the reader has some familiarity with basic calculus, linear algebra, and probability. Any more advanced mathematical tools required will be introduced along the way. Some exposure to game theory is also assumed, though we make minimal use of its formal language.

For a taste of the topics, here are some of the questions we will analyze:

- Disagreement in beliefs can be caused by the parties having different information. Thus the disagreement itself should cause the parties to revise their own beliefs. To what extent can such revision happen? Is it possible to "agree to disagree"?
- The same good can be auctioned off in multiple ways. One can, for instance, make the high bidder pay her bid (a **first price auction**), or ask each bidder to pay her own bid (an **all pay auction**). Which auction format maximizes profit? It turns out that under some conditions, all sensible auctions are the same.
- How do the different electoral systems compare, and how should one choose which to use?
- To what extent will an agent communicate her information to a decision maker? There will of course be full information revelation when the parties have perfectly aligned interests and no revelation when they have fully opposing interests (a **zero-sum game**), but what happens when the parties have imperfectly aligned preferences?
- To what extent can agents' information change the outcome of a game?
- How should a firm optimally design a contract to induce a worker to exert the right amount of effort? What if the firm cannot observe the worker's action (but only their output)? What if the firm doesn't even know what actions the worker has?

And here are some buzzwords: game theory, mechanism design, information economics, auction theory, contract theory, information design, strategic communication, robust mechanism design.

1.1 Notation

We use $\mathbb{R}_+$ to denote the set of nonnegative real numbers, and $\mathbb{R}_{++}$ for the set of strictly positive real numbers.

1. Additional mathematical details will sometimes appear in the footnote.

2 Common Knowledge

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2.1 Aumann's Agreement Theorem

If differences in beliefs originate from asymmetric information, then knowing that others hold different beliefs should make a rational agent revise her own belief. To what extent would this revision happen? Aumann (1976) shows that if agents with different information were to "agree" on their beliefs —[1] if their beliefs are common knowledge — then their beliefs must be exactly the same; agents cannot agree to disagree.

To describe the epistemic model Aumann worked with, we start with the finite state space Ω, describing the set of ex ante possible worlds the agents could be living in. Each **state** $\omega \in \Omega$ is a complete description of a possible world, including the player's beliefs, and the way by which they obtained their beliefs. For players $i = 1, 2$, let $\mathcal{P}_i$ be a partition of Ω that reflects how finely each player can distinguish the different potential states of the world using their observation. For a state $\omega \in \Omega$, write $\mathcal{P}_i(\omega)$ for the cell in $\mathcal{P}_i$ containing ω. If the true state is ω, player i would know that the state must be in $\mathcal{P}_i(\omega)$, but cannot pin down the exact state if $\mathcal{P}_i(\omega)$ contains multiple elements.

Since each ω offers a complete description of the way information is distributed to each individual, we can assume that each agent i knows the partitions of both players $\mathcal{P}_i$ and $\mathcal{P}_j$, though of course she can observe only $\mathcal{P}_i(\omega)$ and not $\mathcal{P}_j(\omega)$ at ω.

This definition is reminiscent of the idea that knowledge is justified true belief. Justified in the sense of concluding from $\omega \in \mathcal{P}_i(\omega) \subset E$, true in the sense that $E \ni \omega$. Gettier cases are not a problem here since in this model we leave no room for epistemic luck. All logical deductions of the agents are always accurate.

Definition 2.1. An **event** E is a subset of the state space Ω. Say i **knows** the event $E \subset \Omega$ at ω if $\mathcal{P}_i(\omega) \subset E$.

To make sense of this definition, consider a proposition q which can be either true or false, and let E denote the set of states in which q is true. When does agent i *know* — believe with certainty — that q is true? She does so exactly when q is true in each of the states deemed possible by her, i.e., when $\mathcal{P}_i(\omega) \subset E$.

The proposition q above can be a statement directly about the state, such as "it is raining today," but it can also be a statement about beliefs, such as "player 1 believes that it is raining today." Given the partition of each player, any statement about beliefs can be translated to an equivalent statement about the state. Statements about beliefs are thus (indirect) statements about the state: When is it true that "player 1 believes that it is raining"? It is true precisely when the state is such that player 1 believes it is raining today. The same is of course true for statements about higher order knowledge.

This sounds like tautology. But the significance of this fact is that using the few ingredients we have, the following definition can be made:

Definition 2.2. Say event $E \subset \Omega$ is **common knowledge** at ω if for any pair of players i and j, i knows E, j knows E, i knows j knows E, j knows i knows E, i knows j knows i knows E, and so on, ad infinitum.

1. All em dashes are human generated.

Write $\mathcal{P}_1 \wedge \mathcal{P}_2$ for the finest common coarsening of $\mathcal{P}_1$ and $\mathcal{P}_2$. We have the following characterization of common knowledge.

The set of all partitions with the refinement order ($\mathcal{P}_1 \leq \mathcal{P}_2$ if $\mathcal{P}_2$ refines $\mathcal{P}_1$) is a lattice. $\mathcal{P}_1 \wedge \mathcal{P}_2$ is the infimum (or the meet) of $\mathcal{P}_1$ and $\mathcal{P}_2$.

Proposition 2.3. *E is common knowledge at ω iff it contains $(\mathcal{P}_1 \wedge \mathcal{P}_2)(\omega)$.*

Proof. At ω, 1 knows that the true state must be in $\mathcal{P}_1(\omega)$. She can thus infer that 2 must have observed $\mathcal{P}_2(\omega')$ for some $\omega' \in \mathcal{P}_1(\omega)$. Thus, 1 knows 2 knows E if for each of these ω', $E \supset \mathcal{P}_2(\omega')$. In other words, 1 knows 2 knows E if E includes all $P \in \mathcal{P}_2$ intersecting $\mathcal{P}_1(\omega)$.

To generalize this idea to higher-order beliefs, say ω' is **reachable** from ω if there is a sequence $P^1, \ldots, P^k$ such that $\omega \in P^1$, $\omega' \in P^k$, and consecutive P^j intersect and belong alternately to $\mathcal{P}_1$ and $\mathcal{P}_2$. The analysis above shows that the kth statement "1 knows 2 knows ... E" holds iff E contains all ω' reachable from ω in step k. Thus E is common knowledge iff it contains every ω' reachable in k steps for $k \geq 0$. This union is precisely $(\mathcal{P}_1 \wedge \mathcal{P}_2)(\omega)$. □

Say an event $E \subset \Omega$ is **self-evident to** i if for each $\omega \in E$ we have $\mathcal{P}_i(\omega) \subset E$. Say E is **self-evident** if it is self-evident to each agent i. It is not hard to see that E is self-evident to i iff it can be written as the union of cells in $\mathcal{P}_i$. Thus, from the previous result, we have the following

Corollary 2.4. *E is common knowledge at ω if there exists a self-evident event F such that $\omega \in F \subset E$.*

Now suppose each player starts with the same prior π over Ω.[2] At ω, upon receiving her information given by $\mathcal{P}_i$, player i updates her belief to $\pi(\cdot \mid \mathcal{P}_i(\omega))$.

Proposition 2.5. *Fix event $A \subset \Omega$ and numbers p_1, p_2. If it is common knowledge at ω that $\pi(A \mid \mathcal{P}_1(\omega)) = p_1$ and $\pi(A \mid \mathcal{P}_2(\omega)) = p_2$, then $p_1 = p_2$.*

Proof. Write $P = (\mathcal{P}_1 \wedge \mathcal{P}_2)(\omega) = \cup_j P^j$ where $P^j \in \mathcal{P}_1$. With the characterization of common knowledge above, we know that $\pi(A \mid P^j) = p_1$ for each j. Thus by the law of iterated expectation, $\pi(A \mid P) = p_1$. A symmetric argument shows $\pi(A \mid P) = p_2$. □

In this sense, it is impossible to "agree to disagree"! So, what do we make of this result when people do seem to agree to disagree?

2.1.1 Is there ever Common knowledge?

The exercise we did takes common knowledge as a starting point, and derives an implication. But the notion of common knowledge as defined by Aumann is strict. Any statement of the form "i knows j knows i knows ..." must be true. Will this ever be satisfied?

Geanakoplos and Polemarchakis (1982) shows that with sufficient communication, the answer is yes. They considered a dynamic version of Aumann's model, with two players starting with possibly different beliefs about the probability of some event A. The players are then assumed to take turns announcing (non-strategically) their assessments of the probability of A, and updating their beliefs suitably (using Bayes rule) after hearing each announcement. Geanakoplos and Polemarchakis (1982) showed that after finitely many rounds of communication, the players' respective assessments will become common knowledge and are thus identical.

But even if common knowledge *can* arise endogenously, *does* it? Perhaps the easier way to induce common knowledge is through public announcements ("town hall meetings"). When information is disseminated in public, not only does everyone know the information, but they also learn that others know this information. But even in such cases, do agents really believe with certainty that all others paid attention and received the information accurately?

Public announcement is a *self-evident* event in the sense defined above.

2. Our model assumes that each player starts from the same prior, which is then updated after she observes the cell containing the true state $\mathcal{P}_i(\omega)$. Hence any differences in priors caused by asymmetric information can be captured in the common prior framework, with players observe different information.

Thus Aumann's result may not always apply in real life, but this doesn't imply that our pursuit has been in vain. We have learned that if people were to truly "agree to disagree," then some assumption in the Aumann's model must fail; some other ingredient needs to be introduced. Perhaps more importantly, we are left with a precisely definition of common knowledge, and a framework for formally reasoning through epistemic statements about what people know, and what they know about what others know.

2.2 Rubinstein's Electronic Mail Game

In cases described above, it is perhaps only believable that agents have "almost common knowledge," where instead of *all* statements of the form "i knows j knows ..." being true, only a (potentially very large) number of these are true. How different is almost common knowledge from common knowledge? Rubinstein (1989) shows that in terms of agents' actions (which is what we ultimately care about), they can be quite different.

Rubinstein considers the following coordination game:

State a

	A	B
A	$\underline{M},\underline{M}$	$0,-L$
B	$-L,0$	$\underline{0},\underline{0}$

State b

	A	B
A	$\underline{0},\underline{0}$	$0,-L$
B	$-L,0$	$\underline{M},\underline{M}$

$$0 < M < L,$$

where states a and b happen with probabilities $1-p$ and $p < 1/2$. The agents in this game would like to coordinate on (A,A) at state a and (B,B) at b. If players coordinate on the action that matches the state, each player gets medium payoff of M. If players coordinate on the action that doesn't match the state, each gets 0, and if players fail to coordinate, the player who plays B gets a large negative payoff of $-L$.

2.2.1 The Complete Information Benchmark

Let's first consider what players would do if the state were common knowledge. In this case, upon knowing the state to be $\theta \in \{a,b\}$, it is as if the agents are both playing the complete information game G_θ, in which payoffs are completely described by the payoffs in the coordination game under state θ. In each state, there are two Nash equilibria: (A,A) and (B,B). In particular, successful coordination — playing (A,A) in a and (B,B) in b — is a Nash equilibrium.

2.2.2 The Electronic Mail

Now suppose only agent 1 observes the state $\theta \in \{a,b\}$. When she sees state b, she immediately sends an email to agent 2. Upon receiving an email, each agent's computer is set up to reply automatically with a confirmation. Due to a technical difficulty, each message will be lost with a probability of $\varepsilon \in (0,1)$ independently. Thus in state b, agent 1 sends an email to 2, who sends a confirmation back to 1, who then confirms the confirmation, and so on until a message is eventually lost (this must happen since $\varepsilon > 0$), at which point each agent's computer displays the number of messages sent by her machine, which we denote as T_i.

Note that T_i counts the "degree" of almost common knowledge in the following sense. Under state a, no messages are sent, $T_1 = T_2 = 0$, and no one knows b (since b is not the state). Under state b, agent 1 knows state b and $T_1 \geq 1$. If $T_2 \geq 1$, then agent 2 received agent 1's first message and thus also both (i) knows state b and (ii) knows that agent 1 knows b. In general, if $T_1, T_2 \geq t$, then all statements of the form "1 knows 2 knows ... b," with no more than t "knows," are true.

What should the players do in this game? How each player plays depends in general on what information they have. In the complete information case, each agent observes the true state and

thus conditions their action on it. When coordinating using the electronic mail, agent 2 knows only how many messages were sent by her machine, T_2; her information is completely described by T_2. While agent 1 observes both T_1 and the true state, her information is also completely described by T_1. This is because she observes state a iff $T_1 = 0$, so from T_1 we can recover all of agent 1's information. In the sequel, we can always view an agent's strategy as a function of T_i.

Let us first consider agent 1's action when $T_1 = 0$, i.e., under state a. To compare the outcome of this game to the successful coordination benchmark, let us suppose that agent 1 plays A in this case. Now, what should agent 2 do when $T_2 = 0$? There are two possibilities for observing $T_2 = 0$. It could be the case that the state is a and thus 1 did not send an email; it could also be the case that the state is b but agent 1's first email is lost. These two cases happen with probabilities proportional to $1 - p$ and $p\varepsilon$. To evaluate the payoffs of playing each action, agent 2 still needs to know what agent 1 would do. In the first case ($T_1 = 0$, state a), we have assumed that agent 1 would play A. In the second case ($T_1 = 1$), it is much harder to say what agent 1 would do since that depends on what agent 2 would do at $T_2 = 1$, which is what we are trying to figure out. However, note that when player 2 plays A, her payoff does not depend on what player 1 does, and when she plays B, we can bound her payoff. This gives the following payoff matrix:

$$\begin{array}{lccc} & \text{prob.} & A & B \\ T_1 = 0 & \propto 1 - p & M & -L \\ T_1 = 1 & \propto p\varepsilon & 0 & \leq M \end{array}$$

Since $1 - p - p\varepsilon > 0$, one can easily show that

$$\text{Payoff from playing } A \;= (1-p)M > -(1-p)L + p\varepsilon M \geq \text{ Payoff from playing } B,$$

so player 2 should play A at $T_2 = 0$.

We can repeat this argument for higher values of T_i, only this time the probabilities and payoffs are slightly different. Let's illustrate with agent 1's decision. When player 1 sees $T_1 = t \geq 1$, two possibilities remain: 1's tth message failed to arrive ($T_2 = t - 1$), or 2's confirmation of this message failed to arrive ($T_2 = t$). These two cases occur with probabilities proportional to ε and $(1 - \varepsilon)\varepsilon$, and in the first case we know by the induction argument that 2 would play A. Using this information, we can construct the payoff matrix as before:

Agent 2's indexing is slightly different (the two possibilities she faces upon seeing $T_2 = t$ are $T_1 = t$ and $T_1 = t+1$), but the argument is exactly the same.

$$\begin{array}{lccc} & \text{prob.} & A & B \\ T_2 = t - 1 & \propto 1 & 0 & -L \\ T_2 = t & \propto 1 - \varepsilon & 0 & \leq M \end{array},$$

which shows that player 1 plays A at $T_1 = t$ as well.

We have now shown the following

Theorem 2.6. *There is only one Nash equilibrium in which player* 1 *plays A in state a. In this equilibrium the players play A independently of the number of messages sent.*

It is in this sense that "almost common knowledge" does not approximate common knowledge: conditional on coordination at state a, with each $\varepsilon > 0$ the agents will not coordinate at b after arbitrarily many messages, even when they both know the state is b. At the limit $\varepsilon = 0$, however, successful coordination at each state becomes possible.

2.3 The Robustness of Equilibria

We have seen now that the widely used common knowledge assumption is highly demanding, and even slight deviations from common knowledge can drastically change the outcome of a

game. So when can we trust the prediction of a game assuming common knowledge? Kajii and Morris (1997) takes a somewhat extreme approach, asking when can the analyst trust the prediction from a complete information game if she is completely agnostic about what higher-order information the agents may have.

Let us fix a complete information game $\mathcal{G} = (I, A, \{g_i\})$ with players $1, \dots, I$, actions $A = \prod_i A_i$, and payoff functions $g_i : A \to \mathbb{R}$. We can augment $\mathcal{G}$ with an epistemic model $(\Omega, \pi, \{\mathcal{P}_i\})$ — where Ω is a finite state space, μ is the common prior, and $\mathcal{P}_i$ is agent i's information partition — to get an incomplete information game $\mathcal{U} = (I, A, \{g_i\}, \Omega, \pi, \{\mathcal{P}_i\})$.[3] In the game $\mathcal{U}$, agents have the same (state independent) utility functions as they do in $\mathcal{G}$.

If all equilibria of $\mathcal{U}$ (for any epistemic model) are close to an equilibrium $\mathcal{G}$, then this equilibrium is robust to informational assumptions. But equilibria in the two games, as described by profiles of strategies, cannot be directly compared. Instead, we will compare the corresponding equilibrium distributions of actions, $\mu \in \Delta(A)$, across the equilibria of both games.

Start by considering the incomplete information game $\mathcal{U}$. We are interested in **Bayes Nash** equilibria of this game, each described by a strategy profile $\{\sigma_i\}$, $\sigma_i : \mathcal{P}_i \to \Delta(A_i)$, such that

$$\sum_{\omega' \in \mathcal{P}_i(\omega)} g_i\left(\sigma(\mathcal{P}(\omega'))\right) \pi\left[\omega' \mid \mathcal{P}_i(\omega)\right] \geq \sum_{\omega' \in \mathcal{P}_i(\omega)} g_i\left(a_i, \sigma_{-i}(\mathcal{P}_{-i}(\omega'))\right) \pi\left[\omega' \mid \mathcal{P}_i(\omega)\right]$$

Note that we abuse notation slightly in writing $g_i(\sigma(\mathcal{P}_i(\omega)))$ for $\sum_a g_i(a)\sigma(\mathcal{P}(\omega) \mid \omega)$.

for each i, $a_i \in A_i$, and each $\omega \in \Omega$. That is, after gaining information $\mathcal{P}_i(\omega)$, agent i's best response is given by the (potentially mixed) strategy σ_i, provided that others also follow σ_{-i}. An alternative viewpoint turns out to be useful: think of an indifferent mediator who privately tells each agent i that the state is contained in $\mathcal{P}_i(\omega)$ and gives an action recommendation following $\sigma_i(\mathcal{P}(\omega))$. The display above then says that each agent will follow the action recommendations of this mediator.

Now, let $\mu \in \Delta(A)$ be the joint distribution of actions corresponding to this equilibrium. It turns out that the following analogous restriction must hold:

$$\sum_{a_{-i} \in A_{-i}} g_i(a_i, a_{-i})\mu(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} g_i(a_i', a_{-i})\mu(a_i, a_{-i}), \quad \forall i, a_i, a_i'. \tag{2.1}$$

Intuitively, if the mediator degrades the information she gives so that instead of telling the agents $\mathcal{P}_i(\omega)$ and giving action recommendations, she only gives each agent the same action recommendations as before, the agents will still have incentives to follow these recommendations.[4] The reverse is also true: when (2.1) is satisfied for some $\mu \in \Delta(A)$, one can find $(\Omega, \mu, \{\mathcal{P}_i\})$ such that the game $\mathcal{U} = (I, A, \{g_i\}, \Omega, \pi\{\mathcal{P}_i\})$ has an (Bayes Nash) equilibrium with action distribution μ: simply construct each agent's partition to reflect the action recommendations for them.

We summarize these results as follows:

Proposition 2.7. *Fix $\mathcal{G} = \{I, A, \{g_i\}\}$. Let $\mu \in \Delta(A)$. Then μ describes the action distribution of some $\mathcal{U} = (I, A, \{g_i\}, \Omega, \pi, \{\mathcal{P}_i\})$ if and only if (2.1) is satisfied.*

3. We additionally (implicitly) assume $\pi(\omega) > 0$ for each $\omega \in \Omega$ so that the posterior beliefs $\pi(\omega \mid \mathcal{P}_i(\omega'))$ are always well-defined.

4. Formally, the defining property of a Bayes Nash equilibrium implies that the function

$$a_i \mapsto \sum_{\omega' \in \mathcal{P}_i(\omega)} g_i\left(a_i, \sigma_{-i}(\mathcal{P}_{-i}(\omega'))\right) \pi\left[\omega' \mid \mathcal{P}_i(\omega)\right]$$

is maximized at each $a_i \in \operatorname{supp} \sigma_i(\cdot \mid \omega)$. Thus whenever a_i is recommended at ω, we have

$$\sum_{\omega' \in \mathcal{P}_i(\omega)} \left[g_i\left(a_i, \sigma_{-i}(\mathcal{P}_{-i}(\omega'))\right) - g_i\left(a_i', \sigma_{-i}(\mathcal{P}_{-i}(\omega'))\right)\right] \pi\left[\omega' \mid \mathcal{P}_i(\omega)\right] \geq 0.$$

Suitably aggregating these expressions gives the desired result.

In this case, we say μ is a **correlated equilibrium** of the *original* (complete information) game $\mathcal{G}$, since it describes the outcome of the game when we allow the agents' actions to be correlated through additional information, or equivalently, the imagined mediator.

We can now answer our original question about robustness:

Corollary 2.8. *If $\mathcal{G} = \{I, A, \{g_i\}\}$ has a unique correlated equilibrium, then this equilibrium is robust to incomplete information in the sense that any $\mathcal{U} = \{I, A, \{g_i\}, \Omega, \pi, \{\mathcal{P}_i\}\}$.*

In fact, more can be said. Suppose the analyst is not just agnostic about the higher-order beliefs, but she is also only confident that the payoffs are correctly specified with high probability. Let us again fix the complete information game $\mathcal{G}$ and consider an incomplete information game $\mathcal{U} = (I, A, \Omega, \pi, \{\mathcal{P}_i\}, \{u_i\})$ with the same number of players and the same action sets as $\mathcal{G}$. Now, player i has utility function $u_i : A \times \Omega \to \mathbb{R}$ depending on both the action profile and the state. We say $\mathcal{U}$ is an **ε-elaboration** of $\mathcal{G}$ if agents have (and know they have) the same payoffs in $\mathcal{U}$ as they do in $\mathcal{G}$ with probability at least $1 - \varepsilon$:

$$\pi\{\omega : u_i(a, \omega') = g_i(a), \forall a \in A, \omega' \in \mathcal{P}_i(\omega), i = 1, \dots I\} \geq 1 - \varepsilon.$$

Denote the max norm on action distributions as $\|\mu - \nu\| = \max_{a \in A} |\mu(a) - \nu(a)|$, and say action distribution μ is **robust to incomplete information** in $\mathcal{G}$ if for each $\delta > 0$ there exists $\varepsilon > 0$ such that all ε-elaborations of $\mathcal{G}$ has an equilibrium action distribution ν with $\|\mu - \nu\| \leq \delta$.

Kajii and Morris (1997) proved the following result:

Proposition 2.9. *If $\mathcal{G}$ has a unique correlated equilibrium μ, then it is the unique robust equilibrium of the game.*

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3 A No-Trade Theorem

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Voluntary trading between individuals is possible only when the agents are different in some way. One agent, for instance, may have more apples than the other (and thus a lower marginal utility for consuming apples), or they might like apples more. Can asymmetric information be the cause of a trade? We will answer this question in the negative following Morris (1994).

The setup of our model is as follows:

- A finite set of states Θ.
- A type space $T = \prod_i T_i$ modelling information.
- Agent $i = 1, \dots, n$. Agent i at type t_i has interim beliefs $\pi_i(t_{-i}, \theta \mid t_i)$.
- Assume each agent is risk neutral and has quasilinear utility.

A trade is a function $\gamma : T \times \Theta \to \mathbb{R}^n$ specifying transfers to each type given the state. Say a trade γ is **feasible** if no outside resource is needed to facilitate it:

$$\sum_i \gamma_i(t, \theta) \leq 0, \quad \forall (t, \theta).$$

Say a trade is **acceptable** if each type has a nonnegative interim payoff

$$\sum_{t_{-i}, \theta} \pi_i(t_{-i}, \theta \mid t_i) \gamma_i(t, \theta) \geq 0, \quad \forall i, t_i,$$

with strict inequality for some i, t_i.

Remark 3.1. The acceptability condition really confused me when I initially studied this model. If each agent only has one *realized* type, why do we care that they would also accept this trade at other types?

One answer is that capturing the outcomes for all possible types (even the ones that can never be realized) is convenient for analysis. To see this, suppose there are two types of player A, t_0^A and t_1^A, and both A and B know that A finds γ acceptable iff she is of type t_1^A. If A is indeed of type t_1^A, she would by assumption find this trade acceptable. However, B does not know what type A has. Accordingly, he evaluates the trade using his interim payoff. In doing so, he analyses whether a trade will be accepted at each possible type profile t and weighs the corresponding payoffs using the probabilities of the types. Instead of going through such trouble, we can simply impose that a trade must be acceptable at all t's — and doing so is without loss, since we can also set the payoffs to be 0 when a trade is not accepted at some profile — so that interim payoffs have the simple form above. ☕

With the setup above, the problem of when there exists a feasible and acceptable trade can be translated to: for which beliefs $\pi_i(\cdot \mid t_i)$ can a set of linear inequalities (those defining feasibility and acceptability) be satisfied? There are well-established mathematical tools deals with such problems. Applying one such tool[1] to our setting yields the following version of no-trade theorem:

Milgrom and Stokey (1982) is perhaps the most famous paper on the No-Trade theorems. Here we instead follow the "no betting case" in Morris (1994), which is more specialized but can be proved more easily with application of Farkas' Lemma. I first learned this result from Ben Brooks, without whom I would never come across this beautiful but little-known paper.

1. See proof details in Chapter A

Theorem 3.2. *There is no feasible and acceptable trade if and only if the interim beliefs $\pi_i(\cdot \mid t_i)$ are consistent with a common prior.*[2]

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2. A subtlety: we cannot establish the fact that the agents actually computed their interim beliefs using the same prior. This is because there are many priors that can lead to the same interim beliefs.

4 Communication

4.1 Cheap Talk

When will communication occur for strategic reasons? Suppose an agent (the Sender, she) has access to some information and can costlessly send an arbitrary message to another agent (the Receiver, he), who then makes a decision. To what extent would the sender reveal her information? When Sender and Receiver has identical preferences, Sender would of course reveal all the information she has, so that Receiver can make an informed decision. When their preferences are completely opposed, Sender would not send any message (or she babbles), since any information will be used by the Receiver, for the Receiver, and thus against the Sender. But what would happen when Sender and Receiver has imperfectly aligned preferences?

It is the assumption that Sender can send arbitrary messages costlessly that makes this model *cheap* talk.

We will analyze this problem using the cheap talk game by Crawford and Sobel (1982), which can be summarized as follows:

- The state θ follows a uniform distribution on $[0, 1]$.
- Sender and Receiver respectively has utility functions $v(a, \theta) = -(a-\theta-b)^2$ and $u(a, \theta) = -(a-\theta)^2$. Thus Receiver tries to match the state, while Sender always prefers the action higher then the state by $b > 0$, which represents her bias.
- Sender observes state θ and chooses a communication strategy $\sigma : \Theta \to \Delta(M)$, where M is the set of possible messages.
- Receiver observes the communication strategy σ and a resulting message m, and then chooses an action $\hat{a}(m)$.

Let's first consider Receiver's strategy for a fixed communication strategy σ. Upon observing a message from Sender, Receiver forms a posterior belief about the state and chooses a to maximize his expected payoff:

$$a \in \arg\max_{a \in A} \mathbb{E}\left[u(a, \theta) | M = m, \text{ where } M \sim \sigma(\cdot|\theta)\right].$$

Because of the quadratic utility specification, his best action would simply be the posterior mean. However, to later check whether the sender has a profitable deviation, we also need to specify what Receiver would do upon observing a message that is never sent under σ, an **off-path** message. In these cases, Receiver cannot form a proper posterior belief (what would you believe upon observing something that should never happen?), and we shall assume Receiver chooses one of the action that he would also choose for an on-path message. This assumption rules out non-credible threats that deter the sender from deviating by threatening to hurt both players.

Before analyzing Sender's strategy, let us consider a few candidate equilibria to gain more intuition. Observe first that a perfectly revealing equilibrium does not exist, for at each state θ, Sender has a profitable unilateral deviation from reporting the truth θ to reporting a slightly higher value, knowing Receiver would then implement the report.

Thus consider the other extreme, a perfectly uninformative equilibrium, where Sender sends the same message independent of the state. Given Sender's strategy, Receiver would gain no

information about the state from the message and thus choose action $a = 1/2$, which is the prior mean of θ. Given our assumption that Receiver would still choose $a = 1/2$ upon receiving any other message from the sender, Sender trivially has no incentive to deviate. Thus the perfectly uninformative equilibrium — called a **babbling equilibrium** — always exists. In this equilibrium, Sender babbles, and no information is conveyed.

Perhaps the most natural intermediate case where only partial information is communicated is that of Sender sending a message telling the Receiver whether θ exceeds some threshold, say θ_1. With this communication strategy and a uniform prior, Receiver would then choose $a = (\theta_1 + 1)/2$ if θ exceeds the threshold and $a = \theta_1/2$ otherwise. To ensure that Sender has no incentives to deviate, it is by continuity necessary for Sender to be indifferent between sending either message when she sees θ_1. This implies

$$v(\theta_1/2, \theta_1) = v((\theta_1 + 1)/2, \theta_1) \iff \frac{\theta_1/2 + (\theta_1 + 1)/2}{2} = \theta_1 + b \iff \theta_1 = \frac{1}{2} - 2b.$$

The first equivalence is shown by noting that $v(\cdot, \theta)$ is symmetric around $\theta + b$. Since Sender's preferred action is increasing in θ, this is also a sufficient condition. Thus, if $b < 1/4$, there exists an equilibrium where Sender reports whether θ exceeds $\theta_1 := 1/2 - 2b$.

The equilibrium above can be termed a partition equilibrium with 2 cells. It turns out that all equilibria in this game are partitional. Given numbers

$$0 = \theta_0 < \theta_1 < \cdots < \theta_N = 1,$$

consider Sender's strategy of reporting m_n whenever $\theta \in [\theta_{n-1}, \theta_n]$. By the same argument as before, we see that this is an equilibrium if and only if

$$v\left(\frac{\theta_{n-1} + \theta_n}{2}, \theta_n\right) = v\left(\frac{\theta_n + \theta_{n+1}}{2}, \theta_n\right) \iff 4b = (\theta_{n+1} - \theta_n) - (\theta_n - \theta_{n-1})$$

for each $n = 1, \ldots, N-1$. Note from the last expression that each interval is longer than the one before it by $4b$, so for fixed b, the number of intervals N is bounded, with the largest possible N decreasing in b. As b gets smaller, partition equilibria with more cells becomes possible. This conforms to the intuition that as Sender and Receiver's preferences gets more aligned, more information can be conveyed.

4.2 Bayesian Persuasion

Kamenica and Gentzkow (2011) asks what happens if Sender is given the power commitment, meaning that she chooses a communication strategy *before* observing the information, and commits to it — sticking with it even when it is in her interest to deviate after observing the state. This is in contrast to the cheap talk model, where we require Sender's disclosure policy to be optimal conditional on her information, no matter what information she might observe; that is, the policy must be *interim* incentive compatible. With commitment, Sender optimizes purely from an ex ante standpoint, knowing that she will follow through.

When is such an assumption realistic? An example is a grading or ranking system: an institution picks and makes public an evaluation procedure before collecting information and has to follow through. Sometimes, commitment is mandated by law, for instances when a prosecutor attempts to persuade a judge by choosing an investigation to be conducted and report its full outcome.

But Sender can also be interpreted metaphorically, as Nature choosing what information about the state is available to Receiver. Solving for Sender's optimal disclosure policy then gives

the best (according to Sender's preference) outcome achievable by varying Receiver's information, and by suitably picking Sender's preference, we can carefully probe the range of possible outcomes achievable under all potential information structures.

Because of this second interpretation, Bayesian persuasion is also referred to as information design. Accordingly, let us switch terminology and call the messages signals, with signal space S, and the communication strategy $\pi : \Theta \to \Delta(S)$ an **information structure**. Note that we can equivalently represent π as a family of distributions over S indexed by $\theta \in \Theta$, i.e., $\{\pi(\cdot|\theta)\}_{\theta\in\Theta}$, or a joint distribution over $\Theta\times S$, $\pi \in \Delta(\Theta\times S)$. The last representation is perhaps most revealing of the fact that Receiver gains information by observing a signal that correlates with the state.

Our new game can be summarized as follows:

- There is a finite set of payoff-relevant states.
- Sender and Receiver respectively have continuous utility functions $v(a, \theta)$ and $u(a, \theta)$, and share a common prior $\mu_0 \in \Delta(\Theta)$ that puts strictly positive probability on each state.[1]
- Sender chooses an information structure $\pi \in \Delta(\Theta \times S)$ for some signal space S.
- Receiver observes the information structure π and a signal realization $s \in S$, and then chooses an action $a \in A$ to maximize his expectation of $u(a, \theta)$.

Let us again first fix an information structure π and a signal realization s, and ask what Receiver would do. Upon observing s, he updates his beliefs using Bayes' rule to $\mu_s \in \Delta(\Theta)$, where

$$\mu_s(\theta) \coloneqq \frac{\pi(s|\theta)\mu_0(\theta)}{\sum_{\theta'\in\Theta} \pi(s|\theta')\mu_0(\theta')}, \quad \forall\theta.$$

With this updated beliefs, he picks the action that maximizes his payoff

$$\hat{a}(\mu_s) \in \arg\max_a \mathbb{E}_{\mu_s}[u(a, \theta)].$$

We write Sender's expected payoff when Receiver has belief μ as

$$\hat{v}(\mu) \coloneqq \mathbb{E}_{\mu}[v(\hat{a}(\mu), \theta)].$$

In particular, note that Receiver's action depends only on his posterior beliefs about the state. These beliefs are stochastic, since with the same π, Receiver can observe potentially different random signals. The distribution of the beliefs, however, are completely determined by the information structure π. Instead of choosing π, let us thus think how Sender would choose a distribution over posteriors directly. Doing so is analytically convenient, since we no longer need to keep track of the space of signals S.

Write such a distribution as $\tau \in \Delta(\Delta(\Theta))$, where we recall that $\Delta(\Theta)$ is the space of beliefs over Θ. We would like to rephrase the problem as maximizing over $\Delta(\Delta(\Theta))$, but not all elements in this space can be induced by some π. For instance, it is not possible for Receiver to have a posterior mean higher than that of the prior after seeing every signal realization — for if he knew he would have beliefs with higher means after getting *any* information, he would have revised his beliefs upwards before getting the information. More generally, his expected posterior belief of a state must agree exactly with his prior belief of that state:

$$\mathbb{E}\,\tau(\theta) = \mu_0(\theta), \quad \forall\theta \in \Theta.$$

In other words, his expected posterior is his prior: $\mathbb{E}\,\tau = \mu_0 \in \Delta(\Theta)$. We say τ is **Bayes plausible** if the condition above is satisfied.

It turns out that Bayes plausibility is the only restriction on τ:

1. Under this assumption, function of the form $\Theta \to \Delta(S)$ can be identified with elements in $\Delta(\Theta \times S)$ with Θ-marginal μ_0, and all posterior are well defined. With Θ finite, there is also no loss in making this assumption since we can always remove states with probability 0 from Θ.

Lemma 4.1. *τ is Bayes plausible iff it can be induced by some information structure π.*

Proof. We have informally argued above for the only if direction. Formally, note that

$$\mathbb{E}[\tau(\theta)] = \mathbb{E}_s[\mu_s(\theta)] = \mathbb{E}[\mu_0(\theta|s)] = \mu_0(\theta)$$

by the law of iterated expectations. For the if direction, set

$$\pi(\mu_s|\theta) := \frac{\mu_s(\theta)\tau(\mu_s)}{\mu_0(\theta)}$$

and observe that π describes an information structure. □

We may now rephrase Sender's problem as

$$\max_{\pi} \mathbb{E}\, v(\hat{a}, \theta) = \max_{\tau : \mathbb{E}\,\tau = \mu_0} \mathbb{E}_\tau\, \hat{v}(\mu).$$

This second statement of the problem leads to a particularly nice geometric interpretation of the problem:

Proposition 4.2. *Denote as* $\operatorname{conv} \hat{v}$ *the convex hull of the graph of $\hat{v}$, and let $V(\mu) := \sup\{z : (\mu, z) \in \operatorname{conv} \hat{v}\}$ be the concave closure of $\hat{v}$. Then, Sender's highest ex ante utility is given by $V(\mu_0)$.*

Proof. Let $G := \{(\mu, \hat{v}(\mu)) : \mu \in \Delta(\Theta)\} \subset \Delta(\Theta) \times \mathbb{R}$ denote the graph of $\hat{v}$. If $(\mu, z) \in \operatorname{conv} G$, then Carathéodory's theorem implies that (μ, z) can be written as a convex combination of finitely many points (μ_i, z_i) in G. Interpreting the weights as probabilities, we see that there is a distribution over these posteriors τ that averages to μ. The corresponding information structure then gives the probability weighted payoffs z_i, which is z. □

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5 Delegation

Aden Chen

Suppose the principal (she) wishes to make a decision $a \in A$, but her optimal decision depends on the state θ which is observed only by the agent (he).

One way for the principal to utilize the agent's information is to directly solicit information from him. Formally, she specifies a mechanism (M, g), where M is a message space and $g : M \to A$ is a decision rule. We restrict attention to deterministic decision rules. The agent conveys information by choosing a message $m \in M$, after which the principal makes decision $g(m)$. Note in particular that the message space M is chosen by the principal, who *solicits* information from the agent, whereas in a cheap-talk game, the message space is chosen by the sender, who *offers* information.

We are assuming that the agent is required to submit a message. This assumption is not crucial for the results.

This game can be described as follows:

- The state $\theta \in [0, 1]$ follows a distribution with density f and cdf F.
- The principal chooses a mechanism (M, g).
- The agent observes state θ and chooses $a \in A$, where $A \subset \mathbb{R}$ is a set of admissible decisions.

We will suppose the principal and the agent respectively has utility functions

$$v(a, \theta) = -(a - a_P(\theta))^2, \quad u(a, \theta) = -(a - \theta)^2,$$

so that the principal's optimal action at θ is $a_P(\theta)$, and the agent's optimal action is θ itself.

Applying the revelation principal, we may restrict attention to direct mechanisms, where messages are states and the decision rule g induces the agent to truthfully report the state.

Lemma 5.1. *The principal's problem is equivalent to*[1]

$$\max_{g:\Theta\to A} \mathbb{E}\left[v(g(\theta), \theta)\right] \quad \text{s.t.} \quad u(g(\theta), \theta) \geq u(g(\theta), \theta), \quad \forall \theta, \theta' \in \Theta.$$

The **incentive compatibility** (IC) constraint above says that the agent must be induced to truthfully report under g. Let's try to deduce some properties of a g that satisfy these constraints. First recall that a type-θ agent's best action is θ itself. This implies that g must be weakly increasing, for otherwise if $\theta_1 < \theta_2$ and $g(\theta_1) > g(\theta_2)$, then one type can pretend to be the other to get a strictly higher payoff. Next, if g is strictly increasing and continuous at $\theta \in (0, 1)$, then type-θ gets his preferred action $g(\theta) = \theta$, for otherwise type θ can nudge the action in either direction by reporting a nearly type in the corresponding direction to obtain a strictly higher payoff. Finally, at a discontinuity point θ, type θ must by continuity be indifferent between $g^-(\theta)$ and $g^+(\theta)$, the left- and right-limits of g at θ. The agent's utility function then implies that $(g^-(\theta) + g^+(\theta))/2 = \theta$. We summarize these findings as follows:

1. Since the agent's report may be mixed (and even while we are assuming deterministic decision rule), the usual revelation principal actually only gives a stochastic direct mechanism $\tilde{g} : \Theta \to \Delta(A)$. However, an argument similar to that used to establish the first part of Proposition 5.2 gives $\sup \operatorname{supp} \tilde{g}(\theta') \leq \inf \operatorname{supp} \tilde{g}(\theta'')$ for $\theta' \leq \theta''$. This shows that replacing $\tilde{g}(\theta) \in \Delta(A)$ by any deterministic $a \in \operatorname{supp} \tilde{g}(\theta)$ preserves the IC constraints. In particular, choosing the principal optimal $a \in \operatorname{supp} \tilde{g}(\theta)$ for each θ gives a deterministic direct mechanism that gives weakly higher principal payoff. It is thus without loss to optimize over these mechanisms.

Proposition 5.2. *Let g satisfy the IC constraints. Then,*

- *g is weakly increasing;*
- *if g is strict increasing and continuous at $\theta \in (0,1)$, then $g(\theta) = \theta$;*
- *if g has a discontinuity at θ, then $(g^-(\theta) + g^+(\theta))/2 = \theta$.*

Another way by which the principal can use the agent's information is to **delegate** her task to the agent. The principal restricts the agents authority by specifying a set of admissible actions $D \subset A$, from which the agent then picks his desired action. It turns out that the delegation problem is equivalent to the information solicitation problem:

Lemma 5.3. *The delegation problem*

$$\max_{D \subset A} \mathbb{E}[v(a_A^\star(\theta), \theta)] \quad s.t. \quad a_A^\star(\theta) \in \arg\max_{a \in D} u(a, \theta)$$

has the same value as the information solicitation problem described in Lemma 5.1.

Note that $a_A^\star(\theta) \equiv a_A^\star(\theta; D)$ specifies the agent's best action in D at θ.

One direction of the equivalence is similar to the revelation principal logic. Take any set D and the associated agent's decision rule $a_A^\star$. The mechanism $g \equiv a_A^\star : \Theta \to D$ is incentive compatible and has the same payoff for the principal. Since we can convert any delegation set to mechanism with weakly higher payoff, mechanisms must perform weakly better than delegation. The other direction can be established similarly, assuming the agent breaks ties in favor of the principal when multiple actions in D solve his problem $\arg\max_{a \in D} u(a, \theta)$. The proof can then be completed by showing that the tie-breaking assumption can be relaxed because any tie-breaking rule results in the same payoff for the principal.[2]

This equivalence means that we can study delegation problem from two perspectives. While looking at delegation sets is perhaps more natural, the mechanism perspective is sometimes more analytically convenient, since it also describes what the agent does at each state.

5.1 When is Delegation Valuable?

Without delegation, the principal's best action is $a_P^\star = \mathbb{E}[a_P(\theta)]$, which can be implemented by giving the agent the delegation set $D := \{a_P^\star\}$. When can the principal strictly improve her payoff by giving the agent more authority? In other words, when does there exist a mechanism that improves upon the best constant mechanism $g \equiv a_P^\star$?

It is natural to first consider a binary mechanism $g_{\theta^\star}$ mapping states lower than $\theta^\star$ to some a_1 and states higher than $\theta^\star$ to $a_2 > a_1$. To make the mechanism incentive compatible, the threshold type $\theta^\star$ has to be indifferent to to the two actions, or $\theta^\star = (a_1 + a_2)/2$. Thus these binary mechanisms look like

$$g_{\theta^\star, \delta}(\theta) := \begin{cases} \theta^\star - \delta, & \theta \le \theta^\star \\ \theta^\star + \delta, & \theta > \theta^\star \end{cases},$$

where $\theta^\star \in (0,1)$ and $\delta > 0$. When can such a mechanism strictly improve upon the constant mechanism $a_P^\star$?

2. To see this, consider any two agent decision rules $g_1, g_2 : \Theta \to D$ for a fixed D. If $g_1(\theta) \neq g_2(\theta)$, then these two actions give the agent the same utility and no other actions in D gives a higher utility. The specification of the agent's utility then implies that the interval from $g_1(\theta)$ to $g_2(\theta)$, of length $|g_1(\theta) - g_2(\theta)|$, does not intersect D. The intervals corresponding to different θ moreover do not intersect because of the previous footnote. The set A being bounded, we know that for each $\delta > 0$ there are finitely many θ such that $|g_1(\theta) - g_2(\theta)| > \delta$. Hence, g_1 and g_2 differ for at most countably many θ and thus gives the same payoff.

We say the principal and the agent are **minimally aligned** if

$$\mathbb{E}[a_P(\theta) \mid \theta \leq \theta^\star] < \theta^\star < \mathbb{E}[a_P(\theta) \mid \theta \geq \theta^\star]$$

for some $\theta^\star$. To get some intuition for this condition, note that the first and last terms are the principal's expected payoff when she knows which side on the threshold $\theta^\star$ the true state lies. Thus we would expect $\mathbb{E}[a_P(\theta) \mid \theta \leq \theta^\star] < \mathbb{E}[a_P(\theta) \mid \theta \geq \theta^\star]$ if the principal and the agent has broadly similar preferences (with ideal points roughly increasing in the state). The condition for minimal alignment additionally impose that the ranges of principal's and agent's ideal actions cannot differ too much, so that the principal's ideal actions in the two cases above cover the agent's preferred action at $\theta^\star$.

When the principal and the agent are minimally aligned, a binary mechanism with the same threshold strictly improves upon the constant mechanism $a_P^\star$. To see this, suppose $\theta^\star \geq a_P^\star$ (the other case is symmetric) and consider the binary mechanism with $\delta \coloneqq \theta^\star - a_P^\star$,

$$g(\theta) \coloneqq \begin{cases} a_P^\star, & \theta \leq \theta^\star \\ \theta^\star + \delta, & \theta > \theta^\star \end{cases}.$$

This mechanism is identical to the constant mechanism $a_P^\star$ when $\theta \leq \theta^\star$ and implements the action $\theta^\star + \delta > a_P^\star$ when $\theta > \theta^\star$, thus giving the principal a higher payoff.

This argument thus shows that minimal alignment is sufficient for delegation to be valuable. It turns out that minimal alignment is also necessary:

Proposition 5.4 (Alonso and Matouschek (2008), Proposition 1)**.** *Delegation is valuable if and only if the principal and the agent are minimally aligned. In particular, delegation is valuable if $a_P(\theta)$ is weakly increasing in θ and $a_P^\star \in (0, 1)$.*

The proof for necessity is more involved, so we skip it.

5.2 Optimal Delegation

Suppose the principal and the agent are minimally aligned, so that delegation is valuable. What should the optimal delegation set look like? With a larger delegation set, the agent can select actions closer to his preferences, which can be further from the principal's ideal point. However, with finer actions choice, the agent can also better put his information to use. Thus we have an information-incentive trade-off.

To be concrete, let's consider the following two delegation sets:

$$D_+ \coloneqq \{a_1, a_2, a_3\}, \quad D_- \coloneqq \{a_1, a_3\},$$

where $a_1 < a_2 < a_3$. Write the midpoint between a_i and a_j as $m_{ij} \coloneqq (a_i + a_j)/2$. The associated decision rules are

$$g_+(\theta) \coloneqq \begin{cases} a_1, & \theta \leq m_{12} \\ a_2, & m_{12} < \theta \leq m_{23} \\ a_3, & m_{23} < \theta \end{cases}, \quad g_-(\theta) \coloneqq \begin{cases} a_1, & \theta \leq m_{13} \\ a_3, & m_{13} < \theta \end{cases}$$

Thus removing the middle action a_2 from D_+ makes agents of intermediate types $\theta \in (m_{12}, m_{13})$ choose the more extreme actions a_1, a_3 instead of the middle action a_2. Whether this benefits the principal depends on her ideal actions function a_P. Intuitively, D_+ performs better when a_P is relatively flat over (m_{12}, m_{13}), and D_- performs better when a_P is steep.

With this intuition, one expects the optimal delegation set to have no holes when the agent has preferences sufficiently aligned with that of the principal. This is indeed the case. Recall that in our current setup, the agent's preferred action at state θ is θ itself. We can interpolate between the principal and the agent's preferred actions to set

$$a_A^\lambda(\theta) \coloneqq \lambda a_P(\theta) + (1-\lambda)\theta, \quad u^\lambda(a,\theta) \coloneqq -\left(a - a_A^\lambda(\theta)\right)^2.$$

An agent with utility function u^λ has preferred action $a_A^\lambda(\theta)$ at state θ, which is close to the principal's preferred action $a_P(\theta)$ when λ is near 1.

Proposition 5.5 (Alonso and Matouschek (2008), Proposition 4)**.** *Suppose a_P is strictly increasing and twice continuously differentiable. Then there exists $\underline{\lambda} \in (0,1)$ such that the optimal delegation set when the agent has utility function u^λ can take the form of a single interval for any $\lambda \geq \underline{\lambda}$.*

We say "can take," since one can always add to a delegation set any action the agent will never choose without changing the principal's payoff. This proposition is thus about the "minimal" optimal delegation set.

5.3 Aligned Delegation

Suppose now that the principal has multiple decisions indexed by $i = 1, \dots, N$. The ith decision concerns state θ_i, which is iid over i. The principal and the agent respectively have utility functions

$$v(a,\theta) \coloneqq -\sum_i (a_i - \theta_i)^2, \quad u(a,\theta) \coloneqq -(a_i - \theta_i - b)^2,$$

where a and θ are action and state profiles. Thus the principal wants to match the state, while the agent has a constant bias b and wants to match $\theta + b$.

The agent observes all states $(\theta_1, \dots, \theta_N) \in \Theta^N$ jointly, makes all decisions $(a_1, \dots, a_N) \in A^N$, and gets a payoff of $u(a,\theta)$. To be more concrete, we may think of the agent as a professor, actions as grades, and states as student qualities. The professor likes to give higher grades if $b > 0$ and lower grades if $b < 0$. The principal, on the other hand, doesn't observe θ_i, and suppose further that she doesn't even know the agent's bias $b \in \mathbb{R}$. Consequently, she chooses a mechanism $\mathcal{D} = (M, g)$ to maximize her minimum utility across all possible values of b,[3] where M is a message space and $g : M \to A^N$ is an action rule.[4]

Such a mechanism is called an **max-min** optimal mechanism.

With multiple decisions, instead of specifying a decision set for each action, the principal has the additional power to restrict the agent's joint action profile, $\{a_i\}_{i=1,\dots,N}$. One such restriction is on the mean action over i, $\bar{a} \coloneqq \sum_i a_i/N$. Call a delegation mechanism with a mean restriction of the form $\bar{a} = \mu$ a **budget mechanism**. Note that

$$u(a,\theta) = -\sum_i (a_i - \theta_i)^2 + 2b\sum_i (a_i - \theta_i) - Nb^2 = v(a,\theta) + 2bN\bar{a} + \text{constant},$$

so with any budget mechanism, the agent (with any b) and the principal solves the same problem: choose a_i to match θ_i while satisfying the mean restriction. Because of this property, we say budget mechanisms are **aligned delegation** mechanisms.

From the expression above, we also see that when b is very large, the term $2bN\bar{a}$ dominates. Thus fixing an arbitrary mechanism $\mathcal{D}$, an agent with a large bias approximately picks actions to maximize $\bar{a}$ as allowed by $\mathcal{D}$. Denote the maximum possible value of $\bar{a}$ allowed by $\mathcal{D}$ as $\bar{a}^\star$

3. Actually, for some mechanisms and some values of b, there might well be multiple equilibria. In these cases we assume the agent plays the principal preferred equilibrium. An economic justification is that a principal with the power to design the delegation rule should also be able to nudge the agent to play any particular equilibrium, since all of them yields the same payoff to the agent. To me this seems like a weak justification especially in a context where we are discussing robustness, but it's a bullet we have to bite.

4. We can allow for much richer mechanisms, including allowing for randomized action rules, or a initial stage where the principal offers a menu of mechanisms from which the principal picks one. The same results hold.

and consider the budget mechanism with mean restriction $\bar{a} = \bar{a}^\star$, which we call $\mathcal{D}^\star$. By the discussion above, an agent with very large b chooses approximately the same actions under $\mathcal{D}$ and $\mathcal{D}^\star$. Also, since $\mathcal{D}^\star$ is an aligned delegation mechanism, it gives the principal the same payoff across all values of b. Thus the worse case payoff of $\mathcal{D}^\star$ weakly improves upon $\mathcal{D}$.

We have shown that the worse case payoff of an arbitrary mechanism $\mathcal{D}$ can be weakly improved upon by some budget mechanism. This implies that when searching for a max-min optimal mechanism, it is without loss to consider only budget mechanisms. Hence:

Proposition 5.6 (Frankel (2014), Proposition 2)**.** *The optimal budget mechanism is max-min optimal.*

Frankel (2014) generalized this logic considerably. A particular case he considered is where players have utility functions with increasing differences

$$u(a'', \theta'') - u(a', \theta'') \geq u(a'', \theta') - u(a', \theta'), \quad \forall a'' > a', \forall \theta'' > \theta',$$

so that the marginal benefit of increasing action is greater in higher states. In this case, when the principal wants to maximize her worse-case payoff, knowing only that the agent has some increasing-differences utility function, Frankel showed that instead of restricting the mean action, the principal wants to restrict every moment of the distribution, or equivalently specify the distribution of actions. Such a mechanism is equivalent to a **ranking mechanism**, where the principal asks the agent to rank each θ_i and implements fixed action rules based on the rankings alone.

5.4 Delegation and Screening

We have been operating under the assumption that the principal only has a limited ability to control actions by restricting the action space. We could instead consider a richer model where the principal chooses a contract specifying transfers to the agent for each action he chooses. Specifically, consider the following setup:

- State $\theta \in \Theta$ follows some distribution, which is common knowledge.
- Finite action space A. The realized state θ is observed privately only by the agent.
- The principal chooses payment rule $t : A \to \mathbb{R} \cup \{\pm\infty\}$ specifying the principal's payment to the agent when action a is chosen.
- The principal has utility $v(a, \theta) - t(a)$.
- The agent has utility $\lambda u(a, \theta) + t(a)$, where $\lambda > 0$ describes how much the agent cares about choosing his preferred right action.

This is typically called a **screening** problem: the agent has some private information which the principal elicits by offering a menu of options designed for the agent to self-select into.

To limit the principal's power, we endow the agent with an outside option with utility normalized to 0 and assume $u \geq 0$ (he could, for instance, decline choosing an action). More specifically, suppose the agent learns the state θ before deciding whether to help the principal. This can be modelled by adding **ex post individual rationality** constraints

$$\max_{a \in A} \lambda u(a, \theta) + t(a) \geq 0, \quad \forall \theta, \lambda \geq 0$$

requiring the agent to be willing to participate in any state θ for any λ.

In this model, delegation is a particular kind of payment rule, one of the form

$$t(a) \coloneqq \begin{cases} 0, & a \in D \\ -\infty, & a \notin D \end{cases},$$

where $D \subset A$ is the set of acceptable actions. It turns out that delegation can emerge endogenously as optimal solutions to screening problems when we add in some form of robustness considerations:

Proposition 5.7. *Suppose both v and u are common knowledge, but only the agent knows λ. The principal chooses t to maximize her guarantee payoff over all possible values of $\lambda \geq 0$. Then, delegation is an optimal solution to the screening problem.*

Proof. Fix any candidate t. Note that at $\lambda = 0$, the agent chooses only $\arg\max t(a)$. Thus the modified transfer rule

$$t'(a) := \begin{cases} 0, & a \in \arg\max t(a) \\ -\infty, & \text{otherwise} \end{cases},$$

gives the principal the same payoff as t when $\lambda = 0$. Moreover, the principal gets the same payoff for each $\lambda \geq 0$ under t', so t' gives weakly higher guarantee than does t. Finally, note that ex post individual rationality for t is equivalent to $\max t(a) \geq 0$, so this modification preserves ex post individual rationality. □

This proof uses the same idea as Proposition 12 in Alonso, Gan, and Hu (2026), but with a slightly different setup.

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6 Contracts

Aden Chen

How should a contract be optimally designed when the relevant actions are not directly observable? More concretely, consider the problem of a firm (the principal, she) incentivizing a worker (the agent, he) to exert effort in their work. The agent's action is not observable and hence **noncontractable**. The principal thus has to write a contract specifying compensations to the worker based (say) on the agent's output, which is observed.

The problem of hidden action is called **moral hazard**, due to the term's initial use in the context of insurance problems, where one might worry that providing insurance gives incentives to being careless or irresponsible. The confusing terminology stuck, but its moral judgement has largely been stripped.

More precisely, something is contractable if it is verifiable by contract enforcing entity, typically the court.

To generate a nontrivial hidden action problem, outputs cannot perfectly reveal the agent's actions. We capture this with stochastic output with distribution determined by the action. Thus we write our base model as follows:

- The principal announces a contract $w : \mathbb{R}_+ \to \mathbb{R}$, which rewards the agent with a payoff of $w(x)$ when his realized output is x.
- The agent decides whether to participate in the contract. He gets utility normalized to 0 from his outside option if he rejects.
- If the agent decides to participate in the contract, he chooses an action $(F, c) \in \mathcal{A}$, where $F \in \Delta\mathbb{R}_+$ describes the stochastic output of the action and $c \geq 0$ is the cost of the action. The set $\mathcal{A}$ describes the technology available to the agent.
- An output x with distribution F is realized, and the agent gets payoff $w(x)$ from the contract.

Suppose the agent has utility function with monetary payoff and action cost additively separable, so that he has ex post utility $u(w(x)) - c$ upon choosing (F, c) and getting payoff $w(x)$ from the contract. The principal is assumed to care only about the agent's output less the payment dictated by contract. For simplicity, we assume that the principal is risk neutral and thus has ex post utility $x - w(x)$. The principal and the agent respectively maximize their ex ante expected utility, $\mathbb{E}_F[x - w(x)]$ and $\mathbb{E}_F[u(w(x))] - c$.

We will first use this model to answer the hidden actions problem posed above following Holmstrom (1979). Then, following Carroll (2015), we will add an additional layer to the hidden actions problem, asking what the principal should do when they are not even sure what actions the agent has.

6.1 Hidden Actions

To see how hidden actions change the contracting problem, we will compare the optimal contract when the action is contractable (known as the **first-best** case) and is noncontractable (the **second-best** case). The symbols $\tilde{w}$ and w are used respectively to denote contracts in the first- and second-best case. Thus in particular, w does not depend on the action (F, c).

First consider the second-best problem. Instead of thinking of the principal as choosing w while anticipating the agent to optimally choose an action, we can equivalently let the principal pick both a contract w and the agent's action (F, c) subject to the constraint that it is in the agent's

interest to choose (F, c):[1]

$$\max_{w,(F,c)\in\mathcal{A}} \mathbb{E}_F[x - w(x)] \quad \text{s.t.} \quad \mathbb{E}_F\left[u(w(x))\right] - c \geq 0,$$
$$\mathbb{E}_F\left[u(w(x))\right] - c \geq \mathbb{E}_{F'}\left[u(w(x))\right] - c', \quad \forall (F', c') \in \mathcal{A}.$$

Here, the first constraint ensures that the agent participates in the contract, and the second constraint restricts (F, c) to be the agent's best action given w. These two constraints are respectively referred to as **individual rationality** (IR) and **incentive compatibility** (IC).

Next, in the first-best problem, the action is contractable, so the principal can induce any action $(F^\star, c^\star) \in \mathcal{A}$ she wants by (say) giving no compensation when any other action is chosen, provided that she gives enough reward to choosing $(F^\star, c^\star)$ that the agent is willing to participate (IR). A contract in this case is

$$\tilde{w}(x, (F, c)) : \mathbb{R}_+ \times \mathcal{A} \to \mathbb{R},$$
$$\tilde{w}(x, (F, c)) \coloneqq 0 \text{ if } (F, c) \neq (F^\star, c^\star),$$

where the principal is left to choose $w(\cdot) \coloneqq \tilde{w}(\cdot, (F^\star, c^\star)) : \mathbb{R}_+ \to \mathbb{R}$, the same object as the second-best case. The first-best problem can now be written as

$$\max_{w,(F^\star,c^\star)\in\mathcal{A}} \mathbb{E}_{F^\star}[x - w(x)] \quad \text{s.t.} \quad \mathbb{E}_{F^\star}\left[u(w(x))\right] - c^\star \geq 0.$$

Thus mathematically, the first-best problem is precisely the second-best problem without the IC constraint!

Next, let us get more concrete and directly compare the first- and second-best *outcomes*. We will first do this comparison under the additional simplifying assumption that the agent has a risk neutral utility function. Later, we will consider risk-averse agents; it is there that our preceding discussion about the structure of the two problems pays off.

6.1.1 Risk Neutral Agents

Let the agent have utility function $u(x) \equiv x$. Observe that with any contract of the form $w(x) = x + \beta$, where β is large enough to satisfy IR, the agent solves

$$\max_{(F,c)\in\mathcal{A}} \mathbb{E}_F[x + \beta - c],$$

which has the same solution as the first-best problem $\max_{(F,c)\in\mathcal{A}} \mathbb{E}_F[x - c]$. The principal optimally chooses the lowest β to barely induce the agent to participate, giving him a surplus of 0:

$$\beta^\star \coloneqq -\left(\max_{(F,c)\in\mathcal{A}} \mathbb{E}_F[x] - c\right) \leq 0.$$

the inequality comes from assuming that the agent has an action generating higher expected payoff than it costs.

Thus the principal can use $w^\star(x) = x + \beta^\star$ to maximize total surplus while leaving the agent with 0 surplus. Since the first-best contract also have these two properties, we see that the principal attains her first-best payoff. Noting moreover that a second-best payoff can never be higher than the first-best payoff (since in the latter case, the principal has more information and hence more contracting power), we find that $w^\star$ contract attains second-best. We summarize this finding in the following proposition:

1. Strictly speaking, this should be a sup that need not be attained without additional assumption. But since such existence problems don't contribute much to the economic understanding of this problem, we shall suppress the relevant technical discussion. In the same vein, we will sometimes refrain from explicitly stating, e.g., standard regularity assumption.

Proposition 6.1. *When the agent has utility function $u(x) = x$, the second-best payoff is identical to that of the first-best and is attained by the contract $w^\star(x)$ defined above.*

There is a natural economic interpretation of this contract: the principal sells the firm to the agent at a price of $-\beta \geq 0$, and the agent gets to keep all of his output.

6.1.2 Risk Averse Agents

Let us now consider the more general case, where the agent has a risk averse utility function u that is increasing and concave. We shall also impose more structure and assume that actions in $\mathcal{A}$ are parameterized by a one dimensional parameter $a \in \mathbb{R}$. Thus each action can be identified with an a, as

$$\Big(F(\cdot \mid a), c(a)\Big) \in \mathcal{A}.$$

The assumption that u is increasing and concave amounts to saying the payoff has positive but decreasing marginal utility for the agent.

Let $f(x \mid a)$ denote the density of $F(\cdot \mid a)$.

As before, we can think of the optimal contract problem as optimally inducing the best action:

$$\begin{aligned} \max_{w(x), a^\star} \ & \mathbb{E}_{F(\cdot \mid a^\star)}\Big[x - w(x)\Big] = \int [x - w(x)] f(x \mid a^\star)\, \mathrm{d}x \\ \text{s.t.} \quad & w \text{ induces the agent to choose } a^\star. \end{aligned}$$

The constraint can again be written more explicitly as

$$\begin{aligned} &\int u(w(x)) f(x \mid a^\star)\, \mathrm{d}x - c(a^\star) \geq 0, \\ &\int u(w(x)) f(x \mid a^\star)\, \mathrm{d}x - c(a^\star) \geq \int u(w(x)) f(x \mid a)\, \mathrm{d}x - c(a), \quad \forall a. \end{aligned}$$

The first and second line respectively ensures the agent participates (individual rationality, IR) and chooses the action desired by the principal (incentive compatibility, IC).

We will now use the **first-order approach**, *assuming* that only local deviations matter in the incentive compatibility constraint. This implies that choosing $a^\star$ under w is locally optimal, thus at $a = a^\star$,

$$\frac{\partial}{\partial a}\Big[\int u(w(x)) f(x \mid a)\, \mathrm{d}x - c(a)\Big] = 0 \implies c'(a^\star) = \int u(w(x)) f_a(x \mid a^\star)\, \mathrm{d}x.$$

Recall that f_a is a shorthand for $\partial f / \partial a$.

We can now write the Lagrangian of the optimal contract problem with IC relaxed to local deviations as

$$\begin{aligned} \mathcal{L} &= \int [x - w(x)] f(x \mid a^\star)\, \mathrm{d}x + \lambda \left(\int u(w(x)) f(x \mid a^\star)\, \mathrm{d}x - c(a^\star)\right) \\ &\quad + \mu \left(\int u(w(x)) f_a(x \mid a^\star)\, \mathrm{d}x - c'(a^\star)\right) \\ &= \int \underbrace{\Big[x - w(x) + \lambda u(w(x))\Big] f(x \mid a^\star) + \mu u(w(x)) f_a(x \mid a^\star)}_{=: I(x; w)}\, \mathrm{d}x - \mu c'(a^\star), \end{aligned}$$

where λ and μ are respectively Lagrange multipliers for the IR and local IC constraints.

Recall again that we are choosing an optimal contract, a function w of realized output x. Thus at the optimum, the derivative of $\mathcal{L}$ with respect to perturbation of $w(x)$ should be 0. Specifically, for any function $h(x)$,[2]

$$\left.\frac{\partial \mathcal{L}(w + \varepsilon h)}{\partial \varepsilon}\right|_{\varepsilon = 0} = \int I'(x; w) h(x)\, \mathrm{d}x = 0,$$

2. This is called a Gateaux derivative, analogous to directional derivatives in calculus.

from which[3]

$$I'(x;w) = -f(x \mid a^\star) + \lambda u'(w(x)) f(x \mid a^\star) + \mu u'(w(x)) f_a(x \mid a^\star) = 0, \quad \forall x.$$

We may rearrange to get the following formula, called a **Borch rule** since it generalizes a related condition derived by Borch (1962) on an earlier risk-sharing problem.

Proposition 6.2. *Assuming the first-order approach is valid, $u' > 0$, and $f(x \mid a^\star) > 0$, we have the following characterization of the optimal contract:*

$$\frac{1}{u'(w(x))} = \lambda + \mu \frac{f_a(x \mid a^\star)}{f(x \mid a^\star)}, \quad \forall x.$$

We can apply $x \mapsto (u')^{-1}(1/x)$ to both sides and get an explicit expression for w, but the characterization above is easy enough to work with and slightly easier to write.

In the formula above, the term $f_a/f = \partial_a \log f$ can be interpreted as local evidence for higher a: the larger is f_a/f, the faster $\log f(x|a)$ increases with a near $a^\star$, and the more likely x is observed under higher a.

Let's compare this result with the first-best case. Recall from previous discussion is mathematically identical to the second-best problem without the IC constraint. Thus with the same analysis as above, we have that the first-best contract is described by

$$\tilde{w}(x,a) = \begin{cases} w^{\mathrm{FB}}(x), & a = a^\star \\ 0, & \text{otherwise} \end{cases}, \qquad \frac{1}{u'(w^{\mathrm{FB}}(x))} = \lambda^{\mathrm{FB}}.$$

If marginal utility u' is strictly decreasing, then the display above implies that the optimal first-best contract is constant, offering full insurance to the agent.

What about the second-best contract? With u is strictly increasing and concave, the function $x \mapsto 1/u'(x)$ is increasing and thus so is its inverse. It turns out that $\mu \geq 0$,[4] so from the relevant Borch rule we see that $w(x)$ is higher when f_a/f is higher, i.e., when there is stronger evidence that the agent exerted more effort. When action is noncontractable, the optimal contract rewards evidence of effort instead of output.

So, is the first order approach actually valid? The answer is not always, as demonstrated by Mirrlees (1999). In fact, it turns to be quite hard to obtain natural sufficient conditions for the validity of the first order approach. Barron (2019) contains more detailed discussion that first introduced me to this issue. Grossman and Hart (1983) presents an alternative analytic framework that avoids the first-order approach.

6.2 Robustness to Unknown Actions

Suppose now that not only are actions hidden and thus noncontractable, the principal is not even certain which actions are available to the agent. The principal knows that the agent's action set $\mathcal{A}$ includes at least a set of known actions $\mathcal{A}_0$, but possibly much more. A natural objective for the principal to maximize here is the **guarantee**, her minimum payoff under all *possible*

3. Technically this equality holds almost everywhere.

4. Denote $s(x) \coloneqq f_a(x \mid a^\star)/f(x \mid a^\star)$. IC restricts the agent's marginal benefit from effort to be $\int u(w(x)) f_a(x \mid a^\star)\,\mathrm{d}x = c'(a^\star) > 0$. Note that the marginal benefit can also be written as

$$\int u(w(x)) f_a(x \mid a^\star)\,\mathrm{d}x = \int u(w(x)) s(x) f(x \mid a^\star)\,\mathrm{d}x = \mathbb{E}\left[u(w(x))s(x)\right] = \mathrm{Cov}\left[u(w(x)), s(x)\right],$$

where the last equality follows from the fact that

$$\mathbb{E}[s(x)] = \int \frac{f_a(x \mid a^\star)}{f(x \mid a^\star)} f(x \mid a^\star)\,\mathrm{d}x = \frac{\partial}{\partial a} \int f(x \mid a^\star)\,\mathrm{d}x = 0.$$

Thus if $\mu \leq 0$, then $u(w(x))$ is decreasing in $s(x)$ by Borch rule, and so $\mathrm{Cov}\left[u(w(x)), s(x)\right] \leq 0$, a contradiction.

action spaces $\mathcal{A} \supset \mathcal{A}_0$. To focus on the problem of unknown actions, we will again impose the simplifying assumption that the agent has a linear utility function $u(x) \equiv x$, and also take away his outside option so he must always participate. To constrain the principal, we additional impose **limited liability**: payoffs to the agent cannot be negative.

The specialized model can be summarized as follows:

- The principal announces a contract $w : \mathbb{R}_+ \to \mathbb{R}_+$ (note that the codomain encodes the limited liability restriction).
- The agent chooses an action $(F, c) \in \mathcal{A} \subset \Delta\mathbb{R}_+ \times \mathbb{R}_+$ and gets utility $\mathbb{E}_F[w(x)] - c$.
- With action space $\mathcal{A}$, the agent thus has utility

$$V_A(w \mid \mathcal{A}) \coloneqq \max_{(F,c)\in\mathcal{A}} \mathbb{E}_F[w(x)] - c.$$

  Similarly, let $V_P(w \mid \mathcal{A}) = \mathbb{E}_{F^\star}[x - w(x)]$ denote the principal's utility, where $(F^\star, c^\star)$ solves the agent's problem above.[5]
- Knowing only $\mathcal{A} \supset \mathcal{A}_0$, the principal maximizes the guarantee $V_P(w) \coloneqq \min_{\mathcal{A}\supset\mathcal{A}_0} V_P(w \mid \mathcal{A})$.

Let us first analyze a linear contract, which is a natural starting point since it induces the agent to care about what the principal cares about (output) in a way that is completely agnostic about actions. Note that for given output x, the principal and agent get $(1-\alpha)x$ and αx under a linear contract $w(x) = \alpha x$. This implies that their ex post payoffs are related via

$$x - w(x) = \frac{1-\alpha}{\alpha} w(x).$$

Now observe that an action (F, c) (possibly outside of $\mathcal{A}_0$) might be chosen by the agent only if $\mathbb{E}_F[w(x)] \geq V_A(w \mid \mathcal{A}_0)$. For any such action, we thus have

$$\mathbb{E}_F[x - w(x)] = \frac{1-\alpha}{\alpha}\,\mathbb{E}_F[w(x)] \geq \frac{1-\alpha}{\alpha} V_A(w \mid \mathcal{A}_0).$$

Applying min over all (F, c) the agent might choose gives

$$V_P(w) \geq \frac{1-\alpha}{\alpha} V_A(w \mid \mathcal{A}_0).$$

Thus if the agent can obtain a positive payoff with known actions, then the principal can guarantee a nontrivial payoff with a linear contract.

It turns out that no other contract can give a higher guarantee, as shown by Carroll (2015). We will not reconstruct Carroll's complete argument (which is exceptionally clear and provides a great deal of intuition; go read it!), but only give a rough summary for some intuition. With any (perhaps nonlinear) contract w, a clever use of a hyperplane separation theorem[6] gives the existence of κ and $\lambda > 0$ such that

$$x - w(x) \geq \kappa + \lambda w(x), \quad V_P(w) = \kappa + \lambda V_A(w \mid \mathcal{A}_0).$$

Note the resemblance of these two formulas with what we derived in the case of linear contracts above.

5. When there are multiple solutions to the agent's problem, assume he breaks ties in favor of the principal.

6. Since we want to derive a relation between the principal and agent's payoffs from the contract (i.e., without factoring in the agent's private cost), it is natural to consider the space of payoff profiles. Fix w and let $\mathcal{S} \subset \mathbb{R}^2$ be the set of all possible payoffs $(\mathbb{E}_F[w(x)], \mathbb{E}_F[x - w(x)])$ from some $F \in \Delta\mathcal{X}$, where $\mathcal{X} \subset \mathbb{R}$ is a compact set of all possible outputs. Let $\mathcal{T} \subset \mathbb{R}^2$ be the set of all payoff profiles (u, v) where $u > V_A(w \mid \mathcal{A}_0)$ and $v < V_P(w)$. The sets $\mathcal{S}$ and $\mathcal{T}$ are disjoint since $v < V_P(w)$ implies that (u, v) cannot be a payoff profile generated by any (F, c) that might be taken by the agent; in particular, it cannot be generated by $(F, 0)$. Thus a hyperplane separation theorem gives κ, λ, μ such that

$$\kappa + \lambda u - \mu v \leq 0, \quad \forall (u, v) \in \mathcal{S},$$
$$\kappa + \lambda u - \mu v \geq 0, \quad \forall (u, v) \in \mathcal{T}.$$

One can then eliminate all possibilities but $\lambda, \mu \geq 0$ and normalize $\mu = 1$.

The formula on the left motivates considering the unique affine contract $\tilde{w}(x) = \alpha x + \beta$ satisfying $x - \tilde{w}(x) = \kappa + \lambda \tilde{w}(x)$. Now, we can apply the same argument we used in the analysis of linear contracts. For any (F, c) that may potentially be chosen by the agent, $\mathbb{E}_F[\tilde{w}(x)] \geq V_A(\tilde{w} \mid \mathcal{A}_0)$, so

$$\begin{aligned}\mathbb{E}_F[x - \tilde{w}(x)] \geq \kappa + \lambda\, \mathbb{E}_F[\tilde{w}(x)] &\geq \kappa + \lambda V_A(\tilde{w} \mid \mathcal{A}_0) \\ &\geq \kappa + \lambda V_A(w \mid \mathcal{A}_0) = V_P(w).\end{aligned}$$

where the second line comes from noting that the agent's payoff weakly improves since $\tilde{w} \geq w$. As before, this implies

$$V_P(\tilde{w}) \geq V_P(w),$$

so the affine contract $\tilde{w}$ weakly increases guarantee.

Now, write $\tilde{w}$ as $\tilde{w} = \alpha x + \beta$ and note that $\beta = \tilde{w}(0) \geq 0$, for payoffs are nonnegative. We can shift $\tilde{w}$ uniformly downwards to $w^\star := \alpha x$, which doesn't change the agent's incentives in choosing her action and weakly improves the principal's payoffs.

We have now shown that the guarantee of any contract w can be improved upon with a linear contract, which implies[7] that linear contracts maximize the guarantee. In fact, one can show under additional assumptions that linear contracts are the only contracts that maximize the guarantee. This model thus provides a new story for the prevalence of linear contracts despite most theoretical models predicting otherwise: the modified Borch rule in general gives a highly nonlinear contract, whereas in the selling-the-firm case gives an affine contract with slope exactly one. The story is perhaps best told through Carroll's own words:

> One way to try to explain [the prevalence of linear contracts] with our model would be to take the model literally, imagining that contract writers explicitly maximize a worst-case objective, are risk-neutral, and so on. A fuzzier story, but perhaps closer to the truth, is as follows. Just as economists work with stylized models for tractability, so, too, real-life decision makers may not be able to write down (or solve) their decision problems in full precision. They may therefore be content to adopt a solution that is guaranteed to perform reasonably well in a class of approximate models …

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7. There is still the technical question of whether an optimal linear contract exists, but let's ignore that.

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A Farkas' Lemma

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Many problems in microeconomic theory (an example is presented in the next section) can be reduced to solving a system of linear inequalities $Ax \leq b$. An useful tool for dealing with systems of linear inequalities is Farkas' Lemma, which gives a necessary and sufficient conditions for the feasibility of a linear system. There are multiple versions of Farkas' Lemma, each suited for different context, but all mathematically equivalent. We shall start with two versions with particularly nice interpretations.

The inequality in $Ax \leq b$ is interpreted componentwise.

Note that for notational clarity, we hereafter follow the tradition of generally not explicitly indicating transposes.

Lemma A.1 (Farkas' Lemma, Geometric version)**.** *Fix $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^m$. Exactly one of the following is true:*

- *There exists $x \in \mathbb{F}^n$, $x \geq 0$ such that $Ax = b$.*
- *There exists $y \in \mathbb{F}^m$ such that $yA \geq 0$ and $yb < 0$.*

Hence Farkas' Lemma is a member of the "theorems of the alternatives," which states that precisely one of a list of conditions must be true.

This says that either b lies in the positive cone generated by columns of A or there is a hyperplane separating this cone from b.[1]

Lemma A.2 (Farkas' Lemma, Algebraic version)**.** [2] *Fix $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^m$. Exactly one of the following is true:*

- *There exists $x \in \mathbb{F}^n$ such that $Ax = b$.*
- *There exists $y \in \mathbb{F}^m$ such that $yA = 0$ and $yb < 0$.*

That is, either the system $Ax = b$ can be solved, or this system is incoherent in the sense that aggregating the equalities with weights y gives a contradiction: $yb < 0 = yA$.

A.1 Specializing Farkas' Lemma for a No-Trade Theorem

We now develop a suitable form of Farkas' lemma that allows us to impose strict inequalities for subsets of constraints, as required by the acceptability property of a trade.

Lemma A.3. *Given $A \in \mathbb{R}^{m \times n}$ and column j, exactly one of the following is true:*

(i) There exists $x \geq 0$, $x_j > 0$, such that $Ax = 0$.

(ii) There exists y such that $yA \geq 0$ with strict inequality for column j.

Proof. Apply Lemma A.1 to $\tilde{A} = [A', e_j']'$ and $\tilde{b} = e_{m+1}$. □

1. Thus one can prove the result above in the case of $\mathbb{F} = \mathbb{R}$ using a hyperplane separation theorem for convex sets, which typically relies on topological properties of the field. However, Farkas' Lemma is true more generally, for arbitrary ordered fields. For those cases, one needs to use an algebraic argument.

2. One can show this version of Farkas' Lemma by applying the geometric version to $\tilde{A} = [A, -A]$ and $\tilde{b} = b$.

Lemma A.4. *Given $A \in \mathbb{R}^{m\times n}$ and a set of columns J, exactly one of the following is true:*

(i) There exists $x \geq 0$, $x_j > 0$ for all $j \in J$, such that $Ax = 0$.

(ii) There exists $j \in J$ and $y \in \mathbb{R}^m$ such that $yA \geq 0$, for all j, with strict inequality for column j.

Proof. From Lemma A.3, (ii) implies not (i). On the other hand, if (ii) does not hold, then Lemma A.3 implies that for each $j \in J$, there exists $x^j \geq 0$ with $x^j_j > 0$, such that $Ax^j = 0$. Then $x \coloneqq \sum_j x^j$ satisfies (i). □

We are now ready to prove the no-trade theorem (Theorem 3.2), which we restate here for completeness:

Theorem A.5. *There is no feasible and acceptable trade if and only if the interim beliefs $\pi_i(\cdot \mid t_i)$ are consistent with a common prior.*

Proof. A trade exists if and only if there are γ_i for each i such that

$$-\sum_i \gamma_i(t,\theta) \geq 0, \quad \forall t,\theta$$
$$\sum_{t_{-i},\theta} \pi_i(t_{-i},\theta \mid t_i)\gamma_i(t,\theta) \geq 0, \quad \forall i, t_i,$$

with the second line being strict for some t_i. This maps into Lemma A.4 case (ii), with rows indexed by $T \times \Theta \times \{1,\dots,n\}$, columns indexed by $(T\times\Theta) \sqcup (\sqcup_i T_i)$, and $J = \sqcup_i T_i$. Thus there is no trade if and only if there exists $\tilde{\phi} : T \times \Theta \to \mathbb{R}_+$ and $\tilde{\lambda}_i : T_i \to \mathbb{R}_{++}$ such that

$$-\tilde{\phi}(t,\theta) + \tilde{\lambda}_i(t_i)\pi_i(t_{-i},\theta \mid t_i) = 0.$$

In this case, $\tilde{\phi} / \sum \tilde{\phi}$ is a common prior. □

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